

Editable and Applicable Indoor Scene Rearrangement via Dynamic Programming

Jinmo Kim , Yunsik Cho, Giri Na, and Jiwon Lee 

Abstract—Scene synthesis is a technology that creates manipulated scenes based on a virtual environment or reality, enhancing visual immersion and facilitating diverse experiences across various domains. This study proposes a new method for arranging objects in an editable structure within indoor scenes using dynamic programming. The proposed method undergoes evaluation and analysis, considering practical application methods and refining synthesis outcomes. The optimal object arrangement approach for indoor scene synthesis via dynamic programming entails a blend of Top-Down problem-solving, wherein objects are interpreted individually based on the scenes' overarching structure and relationships (prior or pairwise) derived from sample scenes, and Bottom-Up problem-solving, which interprets the scenes' structure from the sub-relationships among object units (accessibility). Leveraging a cost function for each method, fast and accurate scene synthesis is achieved through an object arrangement process comprising selection, exploration, convergence, and move. To assess the performance and efficiency of the proposed method, scene synthesis outcomes were generated for eight distinct configurations, with average time and accuracy during the arrangement process evaluated. Furthermore, we verified the flexible editing process of object arrangement by considering user awareness and interaction. Additionally, comparative experiments with existing optimal solution techniques and deep learning-based methods substantiate that the proposed method synthesizes various indoor scenes more efficiently and accurately. Lastly, a user study targeting expert groups was conducted to validate the practical applicability of the proposed method in interior design and home styling.

Index Terms—Scene synthesis, dynamic programming, object arrangement, digital twin.

I. INTRODUCTION

EXTENDED Reality (XR) technology, encompassing Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), advances alongside digital twin technology, facilitating digital modeling, simulation, and monitoring of physical

objects or systems. The Metaverse, an extension of reality into a digital-based virtual world, allowing all activities to occur in an expanded virtual space, is widely utilized across various economic, social, and cultural domains based on the foundations of XR and digital twin technologies [1], [2], [3]. Virtual-scene generation involves creating manipulated scenes based on either non-existent environments or manipulated versions of reality. Duan et al. [4] implemented a university campus and proposed a metaverse architecture encompassing infrastructure, interactions, and ecosystems. Furthermore, research is being advanced by utilizing related technologies such as computer graphics and artificial intelligence. This digital transformation extends beyond landscapes, cities, and architecture to include indoor scenes utilized in fields like interior design, education, and real estate.

Scene synthesis research has primarily focused on structured approaches for scene creation based on rule-based methods and procedural generation. Researchers have explored methods for automatically generating complex and uniquely featured outdoor scenes, such as urban spaces, cities, and landscapes, using procedural models [5], [6], [7], [8]. Indoor scenes require consideration of elements, like furniture arrangements, layouts, and movement paths during composition. Research has explored global optimization techniques using probabilistic meta-algorithms and physics-based simulations [9]. Additionally, deep learning-based methods, such as GRAINS [10], ATISS [11], Scene Mover [12], and HAISSOR [13], have been employed for realistic scene interpolation and completion. To address the limitations of automatic scene synthesis and reduce manual effort in graphic software or editor-type engines, researchers have investigated interactive editing techniques such as MageAdd [14] and Scenedirector [15] for synthesizing scenes with minimal input. However, there are still limitations such as inaccurate scene synthesis results due to local minima in global optimization problems or the lengthy time required for result synthesis. These methods tend to generate synthesis results that heavily rely on the training data, thus limiting the variety and applicability of the results. Achieving scenes tailored to diverse scenarios and concepts requires additional complex and labor-intensive tasks, such as building new training datasets and modifying training models, which can pose challenges for practical applications. Moreover, current methods focus primarily on the final synthetic results rather than the ability to intuitively visualize intermediate steps such as arrangement or structuring. This limitation becomes evident in the absence of a structure that allows users to intervene, edit, or manipulate as needed during the process.

Received 26 August 2025; revised 23 March 2026; accepted 16 June 2026. Date of publication 18 June 2026; date of current version 3 August 2026. This work was supported by Hansung University. (Corresponding author: Jinmo Kim.)

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Digital Object Identifier 10.1109/TVCG.2026.3705434

The aim of this study is to conduct research on scene synthesis, focusing on quickly and intuitively arranging objects in indoor scenes suitable for various situations and purposes. Additionally, experiments were conducted on reality-based virtual scene synthesis, considering its applicability to collaborative XR environments using MR based on a digital twin. To achieve this, we propose a novel indoor scene synthesis method using dynamic programming, which solves complex problems by breaking them into simpler subproblems. Accurate results are efficiently calculated in intuitive structure compared to existing optimal solution methods by aligning the overlapping subproblems and optimal substructure conditions of dynamic programming with the object arrangement process in indoor scenes. Additionally, we introduce a novel reality-based sample automation-based rearrangement, addressing limitations of dependence on samples and sample constraints, facilitating practical applications of scene synthesis results. This applied approach enables the reorganization and improvement of indoor spaces to suit specific situations and purposes based on real-world data. The combination of scene understanding, custom vision, and scene synthesis algorithms holds significant potential for advancing scene synthesis research. Therefore, another key objective is to scan real scenes and restructure them into virtual scenes in a similar manner, thereby enabling various applications.

The contributions of this study are as follows. First, by leveraging dynamic programming algorithm design techniques, the position and orientation of the object arrangement were interpreted on an object-by-object basis through partial problem repetition. This efficient scene interpretation and arrangement can be divided into Top-Down and Bottom-Up approaches.

1. **Editable:** Instead of calculating all objects in the scene simultaneously, the synthesis of the entire scene is viewed as the optimal substructure of each object unit. This enables the arrangement process to be intuitively checked, facilitating user editing and intervention by designing a structure that supports it.
2. **Applicable:** In contrast to existing research that solely focuses on scene synthesis outcomes, our study explores reality-based scene synthesis application research for digital twin-based collaborative XR. We consider practical applications across various industrial fields.

The efficiency of the proposed method is objectively analyzed through quantitative comparative experiments with existing optimal solution and interactive editing studies. In addition, the research methodological qualitative comparison with recent research trends in deep learning and LLM (Large Language Model)-based scene synthesis is performed to clearly present the differentiation, applicability, and strengths of this study. Finally, for novel applicability different from existing scene synthesis studies, various rearrangement results are generated through reality-based scenes. And a clear direction is examined through statistical analysis through expert surveys.

II. RELATED WORK

The development of scene creation methods, essential for understanding specific situations, places, and backgrounds in

virtual environments, thereby providing new visual experiences. This technology is crucial alongside the advancement of digital twins and metaverse industries based on XR. Several studies have been conducted on generating outdoor scenes, including research on cities, buildings, landscapes, architecture, and façades. Several approaches, such as procedural modeling [8], shape grammar [16], and image-based modeling [17], have been explored. Vanegas et al. [5] proposed a framework that adds intuitive and high-level control to existing urban procedural models. Additionally, a virtual landscape with numerous trees was created based on growth rules, grammar, and mathematical functions [6]. Furthermore, research has explored the automatic creation of virtual scenes using maze-creation algorithms for themed mazes [7].

Unlike outdoor spaces, designing indoor scenes often requires algorithms or learning models to understand the structure and relationships, including the arrangement, layout, and movement lines of objects (furniture, props, etc.) that make up the scene. Techniques for indoor scene synthesis involve composing new scenes using optimization algorithms based on sampling methods. These methods obtain random samples from probability distributions, considering predefined information, relationships, and structures of the objects constituting the virtual scene (or new objects by category). These methods aim to construct novel scenes distinct from existing ones. In this regard, existing studies have investigated rational indoor scene synthesis using geometric approaches to predict furniture arrangements from a semantic perspective or to create realistic scenes. Yu et al. [9] proposed an optimized object arrangement method for natural scene synthesis, applying the Markov chain Monte Carlo (MCMC)-based Metropolis-Hastings sampling technique. Weiss et al. [18] applied an elastic simulation based on position-based dynamics to scene synthesis, demonstrating that large-scale outdoor scenes can be synthesized quickly compared to existing research based on MCMC. Yeh et al. [19] implemented an optimal configuration for two open-world layout synthesis problems—a coffee shop and golf course—using a new MCMC algorithm. Additionally, research has been conducted on designing real scenes using transformable furniture layouts [20]. Kan and Kaufmann [21] proposed greedy cost minimization as a novel method for rapidly generating furniture arrangements for indoor scenes. Another approach involves constructing prior probabilities directly from samples in the dataset, rather than sampling from probabilistic distributions. Furthermore, there are studies on geometry-based layout-generation algorithms that aim to express the relationships between three or more objects which can be challenging to model mathematically [22].

With the advent of deep learning, machine-learning-based approaches have been used in scene synthesis research [23]. Techniques such as feed-forward neural networks, generative adversarial networks, and variational autoencoders enable the creation of realistic scenes through artificial neural network models. The interpolation and completion of scenes resembling reality, matching various situations, were achieved using a graph neural network [24], [25], [26], [27], [28]. Kim et al. [29] introduced a new approach to scene generation using a hierarchical latent diffusion model. Additionally, SceneHGN

was proposed [30], capable of generating plausible 3D room content, including furniture and layouts with fine-grained geometry. Another approach, Mining Interaction and Movement to infer 3D Environments, is a generative model that generates 3D indoor scenes based on 3D human movements [31]. Finally, HAISOR, a scene reconstruction algorithm based on reinforcement learning that considers user interactivity, has been recently announced [13]. Recently, research has been conducted to enable the creation of various scenes by intuitively reflecting the user's intention into scene synthesis through LLM-based prompt engineering. LayoutGPT by Feng et al. [32] applied LLM to 3D indoor scene synthesis by generating layouts from text conditions and increasing numerical and spatial accuracy through the role of visual planners. HOLODECK proposed by Yang et al. [33] is a system that fully automatically generates a 3D environment according to the user-supplied prompt. It can generate various scenes such as arcades, spas, and museums, and can also give meaning to complex queries. DiffuScene by Tang et al. [34] is basically a scene synthesis research based on deep learning, but it also presents a scene synthesis support function that reflects the input text prompt by utilizing BERT (Bidirectional Encoder Representations from Transformers)-based text embedding.

However, most studies on scene synthesis do not allow user editing and manipulation during the process. To address this limitation, various scene synthesis tools have been proposed, including those that incorporate the subjective opinions of designers using semiautomatic object recommendation algorithms [14] as well as intelligent scene synthesis tools that allow designers to quickly understand potential synthesis results [15]. However, existing studies have primarily focused on synthesis results, analyzing the overall structure and relationships of the scene. This approach makes it difficult to intuitively check the scene synthesis process and contributes to time inefficiency. This paper proposes a new scene synthesis method that applies dynamic programming to efficiently synthesize scenes. The method interprets the overall structure of the scene and the relationships and information of object units separately.

Studies on creating scenes based on reality have explored instantiating virtual scenes using physical spatial information from the real world. Cheng et al. [35] introduced VRoamer, a system that procedurally generates unseen physical spaces based on real-world data for VR users. Similarly, Qian et al. [36] restructured real scenes into virtual environments using Microsoft HoloLens 2 and Azure Custom Vision. They provided virtual environment editing functions for VR users to create content for AR users based on this reconstruction. Oasis, which calculates the walkable area based on the scanned 3D real space and rules for creating a virtual environment, has also been proposed, allowing multiple users to interact smoothly in a shared virtual environment [37]. Recently, a solution was introduced to address real-world obstacle-avoidance problems by rearranging the virtual scene based on the user's physical space [38]. These studies are advancing towards 3D reconstruction studies of task spaces aimed at improving MR remote collaboration [39]. Ultimately, seamless collaboration between real-world users and VR users based on digital twins requires technology to quickly and

intuitively implement scenes with various compositions tailored to different situations and purposes alongside reality-based virtual scene synthesis. From this perspective, this study proposes directions for the application of reality-based virtual scene synthesis technology, alongside efficient scene synthesis techniques.

III. DYNAMIC PROGRAMMING BASED SCENE SYNTHESIS

Dynamic programming [40] is an algorithm design technique that breaks down a complex problem into smaller subproblems, solves each problem, and derives the solution to the original problem based on these subproblems. This approach simplifies solving overlapping subproblems, where identical issues recur, and problems with an optimal substructure, allowing the overall solution to be derived from optimal partial solutions. For natural and optimized object arrangement for indoor scene synthesis, numerous factors must be considered, including the overall structure of the scene, relationships between objects, and potential overlap between objects. Approaching the problem from the perspective of dynamic programming, object-object relationships or overlaps can be interpreted as overlapping subproblems. For example, calculating the overlap between a desk and a closet is analogous to calculating the overlap of a closet based on a desk, and vice versa. Once optimal positions and orientations for each object in the scene are determined, the natural arrangement of objects for the entire scene is achieved, resulting in optimal substructure. Therefore, this study proposes a new optimal solution method for Top-Down and Bottom-Up problem solving to facilitate intuitive object arrangements for indoor scene synthesis. Fig. 1 summarizes the overall structure of the proposed method based on dynamic programming.

A. Object Representation

Object arrangement through dynamic programming involves Top-Down and Bottom-Up problem-solving methods, with object representation defined for optimal solution calculation using a cost function. This study utilized simple spatial relationships regarding the layout or pairwise relationships based on the bounding surfaces of objects [9]. Based on the basic properties of the i -th object (center (p_i), surface length (b_i)), the structure between the object and layout (prior relationship: distance between the nearest walls (d_i) and orientation (θ_i)) from the sample and the relationship between the objects (pairwise relationship: distance between objects (d_i), direction relationship relative to the parent object (θ_i), and position relationship (ϕ_i)) (Fig. 2).

B. Top-Down Approach

The Top-Down problem solving method involves breaking down a large problem into smaller subproblems. In this study, the proposed Top-Down approach interprets the structure or relationships of each object (smaller subproblems) from samples with the structure of the entire scene (large problem). The spatial relationships considered included a prior and pairwise relationships. The prior relationship represents the ideal transformation (p, θ) of an object within the overall scene layout, whereas the

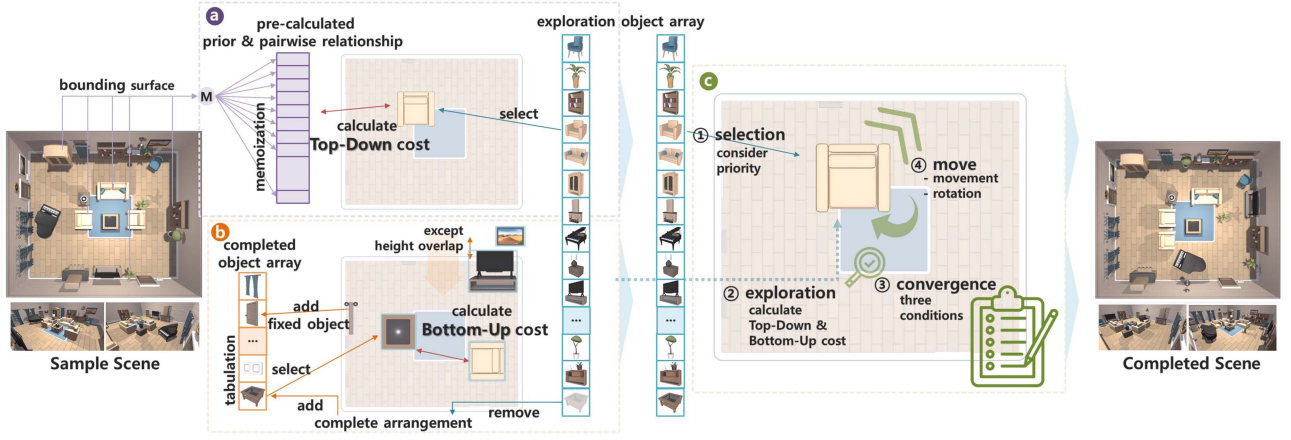


Fig. 1. Structure of object arrangement based on the proposed dynamic programming method in indoor scene synthesis; (a) Top-Down approach, (b) Bottom-Up approach, (c) object arrangement process.

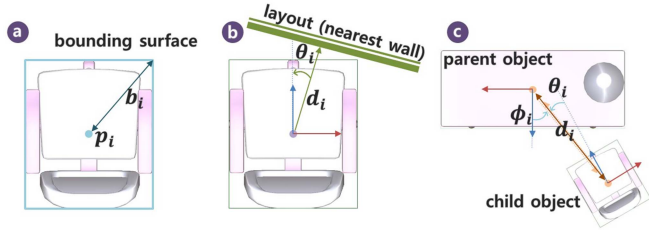


Fig. 2. Object representation: (a) bounding surface, (b) prior relationship with layout, and (c) pairwise relationship.

pairwise relationship describes the mutual relationship between objects. These relationships were calculated and stored in advance from sample scenes. Similar to memoization in dynamic programming, it is based on a structure that caches previously calculated values and reuses them when needed. At this point, the size of the memoization is proportional to the scale of the sample scene (such as the number of objects and relationships, Fig. 2(b) and (c)). The cost values (C_d^{pr} , C_θ^{pr} , C_d^{pair} , C_θ^{pair} , and C_ϕ^{pair}) of the current object are calculated using the Euclidean distance from the memoized prior and pairwise relationship values from the sample scene (Fig. 2).

Based on each calculated cost value, the Top-Down cost function is given by (1):

$$\begin{aligned} C_d^{td} &= w_d^{pr} C_d^{pr} + w_d^{pair} C_d^{pair}, \\ C_r^{td} &= w_\theta^{pr} C_\theta^{pr} + w_\theta^{pair} C_\theta^{pair} + w_\phi^{pair} C_\phi^{pair} \end{aligned} \quad (1)$$

Here, w_d^{pr} , w_d^{pair} , w_θ^{pr} , w_θ^{pair} and w_ϕ^{pair} are weights for each cost and are defined as values between 0.01 and 0.1. However, for child objects in a pairwise relationship, the relative relationship based on the parent is more important than the prior relationship from the sample scene. Therefore, w_d^{pr} and w_θ^{pr} are set to 0 (Fig. 1(a)).

C. Bottom-Up Approach

Bottom-Up problem-solving involves starting from small subproblems, storing the results, and gradually solving larger problems. Accessibility is a spatial relationship interpreted on an object unit. Previous research on indoor scene synthesis, often involves computational redundancy due to the calculation of costs and formulas for entire objects. In this study, we approach the arrangement and relationships in indoor scenes from a cognitive perspective, focusing on large objects (focal objects, objects of high importance, etc.) when considering the structure and arrangement of indoor scenes, such as interior design and home styling. That is, we arrange objects sequentially, starting from one object (small subproblems). Equation (2) represents the cost value for accessibility and the Bottom-Up cost function based on it.

$$\begin{aligned} C_{ac} &= \sum_i \sum_j \max \left[0, 1 - \frac{\|p_i - p_j\|}{b_i + b_j} \right], \text{ if } i \neq j \\ C_a^{bu} &= w_{ac} C_{ac} \end{aligned} \quad (2)$$

C_{ac} represents the accessibility of objects in the completed object array relative to the current object. The final Bottom-Up cost is calculated by accumulating these accessibility values. The parameter w_{ac} serves as a weight and is adjusted within the range of 0.05 and 0.1.

Once the transform (p, θ) of the current object is determined, a tabulation process is initiated to sequentially fill the array with completed object information. Accessibility for the next object minimizes unnecessary operations by comparing and calculating only the objects stored in the completed object array. This demonstrates that the efficient calculation of accessibility through a table-filling arrangement that stores completed objects sequentially (Fig. 1(b)). Additionally, fixed objects such as doors, windows, and built-in closets, which do not require arrangement due to their fixed positions in the scene, are defined. These objects are stored in the completed object array at the beginning, excluded from the arrangement process, and only

used to calculate Bottom-Up costs. This approach contributes to the rapid and accurate creation of scenes.

One exception to the accessibility calculation occurs between the current object and the completed object array. Although indoor scenes exist in three-dimensional space, the calculation of Top-Down and Bottom-Up costs occurs in the x-z plane, representing a two-dimensional space. This approach can lead to a problem of height overlap in accessibility. Height overlap may be mistakenly identified as overlap in the x-z plane, negatively impacting the generation of diverse and natural scene synthesis results. To address this issue, exceptional situations without height overlap are identified, and an exception condition is added to exclude them from the Bottom-Up cost calculation (Fig. 1(b)).

D. Arrangement

1) *Selection*: Based on the object representation, the arrangement of all objects constituting the scene is performed by calculating a cost function divided into Top-Down and Bottom-Up approaches. The proposed object arrangement process consists of four steps: selection, exploration, convergence, and move. The object arrangement proceeded sequentially. Object arrangements based on dynamic programming consist of two arrays: the exploration object array, comprising objects yet to be arranged, and the completed object array, containing objects that have been arranged. The arrangement begins by selecting one object from the exploration object array.

Regarding the arrangement order, existing studies utilize deep learning to arrange objects one by one in an autoregressive manner. In addition, the general order is defined using the spatial arrangement of objects or class frequency, which can cause unintended problems, such as struggling to create more general objects after fewer general objects [11]. This study proposes a method that can apply randomness and diversity to object selection (arrangement order) using the feature of interpreting a scene by an object unit.

This study divided object selection into two types: random and priority-based. Random selection selects one of the objects in the exploration array with equal probability. However, child objects in pairwise relationships are arranged based on their parent objects; therefore, they are excluded from random search. The followings are the priority-based selection, which is a representative priorities in this study.

1. pairwise relationship: parent object with many child objects
2. pairwise relationship: child object in the process of arrangement
3. none relationship: objects with a large bounding surface (b_i)

Objects with a pairwise relationship (e.g., TV-sofa, dining table-stool, desk-office chair) are arranged in the scene as a pair. When arranging a parent object (table), the child objects (chairs) calculate their transformation (position, orientation) based on the parent. Therefore, objects with pairwise relationships must consider a relatively large arrangement area to ensure appropriate positioning of the child objects. Otherwise, they may converge to a local minimum where the child object cannot

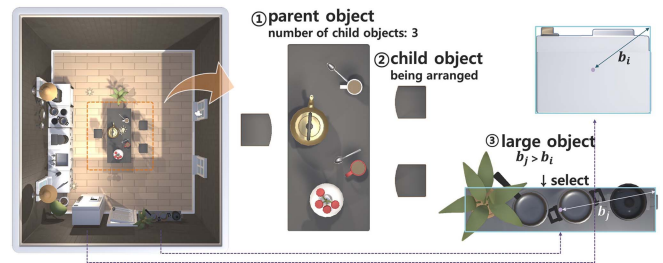


Fig. 3. Example of object selection considering priority from exploration object array.

find a suitable location for arrangement. Hence, the priority is determined based on the number of child objects among the parent objects in the pairwise relationship. Subsequently, after arranging the parent object, the associated child object is selected, and the arrangement is executed based on the parent's positioning. Once all child objects are arranged, the arrangement process for the current pairwise relationship concludes. In cases where no pairwise relationship objects exist, the space is filled initially with larger objects (Fig. 3). However, this presents only one possibility, as users can prioritize objects that they deem important.

2) *Exploration*: After selecting the object for arrangement, the next step is the exploration process. This involves calculating the Top-Down (C_d^{td} , C_r^{td}) and Bottom-Up (C_a^{bu}) costs for the current object. Additionally, for child objects involved in pairwise relationships, a single rotation component (ϕ_i) (Fig. 2(c)) is considered. This includes not only the basic properties such as position (p_i) and orientation (θ_i), but also the relative position (ϕ_i) based on the parent.

3) *Convergence*: Once each cost value of the current object is calculated through the exploration process, arrangement completion is determined through the convergence condition. The convergence conditions are divided into the following three conditions:

1. ordinary object without a pairwise relationship or parent object with a pairwise relationship
2. child object of a pairwise relationship
3. probability conditions for convergence

First, the convergence conditions are checked to ensure that realism is preserved, avoiding overlap for all objects. This involves checking whether all three conditions converge within the threshold for the Bottom-Up cost (C_a^{bu}). In the first case, convergence is checked based on the threshold for the Top-Down cost (C_d^{td} , C_r^{td}). The second condition pertains to a child object of pairwise relationships, where convergence is assessed through the cost, including the relative position based on the parent object. Here, the threshold determines the balance between accuracy and arrangement speed (time). A wider range facilitates faster convergence but may lead to inaccuracies or unnatural arrangements. Conversely, a tighter range may prolong convergence or prevent it altogether. Therefore, this study introduces a probability condition that relaxes the constraint on distance (d) under the premise that the Bottom-Up cost must necessarily hold. Equation (3) represents the process of adjusting the

distance cost based on the number of repetitions of each object arrangement.

$$C_i^{prob} = C_{d_i}^{td} \cos \left(\min \left(\frac{\pi}{2}, \frac{\pi}{2} \times \frac{t}{t_{max}} \right) \right) \quad (3)$$

C_i^{prob} represents the probability cost adjusted based on the number of arrangement iterations (t). If convergence is not achieved even after repeated arrangement processes up to t_{max} (e.g., 5,000~10,000 times), the distance cost is excluded from the convergence conditions. In other words, this introduces a probability to converge only if the rotation cost is satisfied.

4) *Movement*: Finally, if the convergence condition is not satisfied, the object arrangement involves both movements and rotations based on the calculated cost values obtained during the exploration process. Movement occurs within a predefined layout, considering the calculated cost and prior (or pairwise) relationships and accessibility between objects. Equation (4) was used for the movement calculation, reflecting a normal distribution to translate objects according to a Top-Down relationship (prior or pairwise). Subsequently, fine adjustments are applied based on accessibility cost. This entails setting the direction of movement from the current position to the next position is set randomly using a normal distribution, with the movement amount weighted proportionally to each Top-Down cost.

$$(p_i^x, p_i^z) = (p_i^x, p_i^z) + \delta p \{ C_d^{td}(w, h) + C_a^{bu}(a^x, a^z) \} \quad (4)$$

(p_i^x, p_i^z) represents the current position of the object, (p_i^x, p_i^z) denotes the position after movement, δp stands for the normal distribution, C_d^{td} is the Top-Down cost of the current object regarding the calculated distance, and C_a^{bu} is the Bottom-Up cost. Additionally, (w, h) denote the width and height values of the layout, while (a^x, a^z) represent the weights associated with accessibility for fine movement.

For rotation, it's essential to differentiate between child objects with pairwise relationships and other objects. If there are no children or if it is a parent object in a pairwise relationship, we consider comparison with the prior relationship memoized from the sample during the Top-Down approach. Equation (5) describes rotation for an object without a pairwise relationship or a parent object in a pairwise relationship. It involves a four-way rotation centered on the current rotation angle. θ_i represents the rotation value of the current object, θ'_i denotes the result after rotation, and a natural number k is randomly selected to perform a four-way rotation.

$$\theta'_i = \theta_i + \frac{\pi}{2} (k \bmod 4), k \in \{0, 1, 2, 3\} \quad (5)$$

However, in the case of a child object of a pairwise relationship, two rotations are considered. The first rotation (θ) determines the direction in which the child object faces its parent. The second rotation (ϕ) determines the relative position of the child object based on the position of the parent object. Firstly, it rotates around itself to find the appropriate direction to face the parent object. Then, it rotates around the parent object to find the appropriate relative position. Equation (6) illustrates the rotation calculation, considering normal distribution and the Top-Down

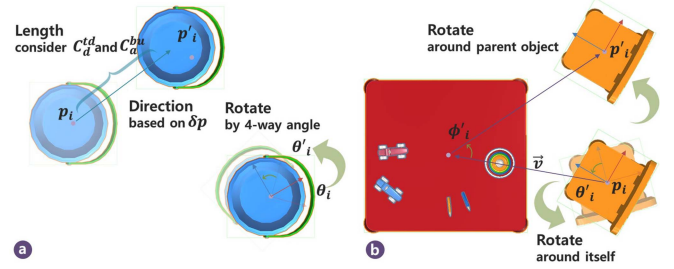


Fig. 4. Example of the proposed object arrangement move process; (a) an object without a pairwise relationship or a parent object of a pairwise relationship, (b) a child object of a pairwise relationship.

cost around itself. θ_i and θ'_i are the same as in (5), $\delta\theta$ represents normal distribution, and C_r^{td} is the Top-Down cost related to rotation.

$$\theta'_i = \theta_i + \delta\theta C_r^{td} \quad (6)$$

Equation (7) depicts the process of calculating the position while rotating around the parent object.

$$\begin{aligned} \vec{v} &= p_{pr} - p_i \\ \phi'_i &= \phi_i + \delta\phi \\ p'_i &= T^{-1}(\vec{v})\phi'_iT(\vec{v}) \end{aligned} \quad (7)$$

Where $\vec{v}$ is a vector pointing from the position of the child object (p_i) to the position of the parent object (p_{pr}). The position of the child object is translated to the position of the parent object using $\vec{v}$, then rotated considering normal distribution. Then, it is translated back to $\vec{v}$ and updates the relative position (p'_i) centered on the parent object. In essence, the child object considers both the directional relationship (θ) and the positional relationship (ϕ) with the parent object. Fig. 4 illustrates movement according to the types of the object.

5) *Process*: Algorithm 1 outlines the complete procedure of object arrangement, encompassing four steps: selection, exploration, convergence, and move. Fig. 1(c) illustrates this step-by-step process.

During the object arrangement process, additional measures may be necessary if a position is calculated that deviates from the layout. The nearest wall decision was applied by calculating the shortest distance between the wall and the object. And, the outside area was calculated through the dot product between the front direction vector of the wall and the vector heading from the wall to the object. Finally, after completing the arrangement, a correction process (fine arrangement) was applied to induce natural results by attaching wall objects (picture frames, closets, etc.) to the nearby wall. Equation (8) shows the fine arrangement for the defined object (e.g., wall object). From the current position of the object (p_i), calculate the final position (p'_i) through the shortest vector ($\vec{v}_p^w$) toward the nearest wall. s_i^f is the half length of the object's bounding surface in the forward direction. This takes into account the different bounding surfaces of each object and calculates the exact position based on the wall without

Algorithm 1: Object Arrangement via Dynamic Programming.

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1: procedure OBJECT ARRANGEMENT
2:   exploration object array  $\leftarrow$  store the entire objects
    that make up the scene.
3:   completed object array  $\leftarrow$  add objects whose
    arrangement has been completed.
4:   fixed objects are added to the completed object array.
5:   select objects based on three conditions defined in
    the exploration object array.
6:   explore for current object.
7:     calculate of  $C_d^{td}$ ,  $C_r^{td}$ ,  $C_a^{bu}$  using (1) and (2)
8:   convergence judgment based on three defined
    conditions.
9:   if convergence then
10:    add the current object to the completed object array
    and remove it from the exploration object array.
11:  else
12:    move on current object (movement, rotation)
    based on (4)-(7)
13:  end if
14:  repeat steps 5-13 until the exploration object array is
    empty.
15: end procedure

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going inside the wall.

$$p'_i = p_i + \frac{v_p^w}{\|v_p^w\|} \left(\|v_p^w\| - s_i^f \right) \quad (8)$$

IV. EDITING

Existing studies are based on probabilistic optimization techniques or arranging all objects that compose a scene simultaneously through a repetitive structure with general sampling. Most studies on scene synthesis using deep learning do not consider interactions in the arrangement inference process other than input conditions or result generation after inference. This study emphasizes the features of sequentially arranging and designing objects such that editing is possible during the arrangement process.

The proposed scene synthesis method using dynamic programming automatically arranges all the objects that comprise an indoor scene. However, there are cases in which the arrangement process converges to a local minimum, such as the probabilistic convergence in this study. This can cause cognitive or aesthetic discomfort in generated scenes. Therefore, this study proposes a method for solving this problem by converting the local minima problem into a probabilistic convergence problem and applying an editing function. Fig. 5 shows an example of the editing process. An unnatural situation may occur when a bookshelf is placed in the middle of a scene because of probabilistic convergence, and the arrangement may continue to proceed. In such cases, this can be a fatal problem during scene synthesis.

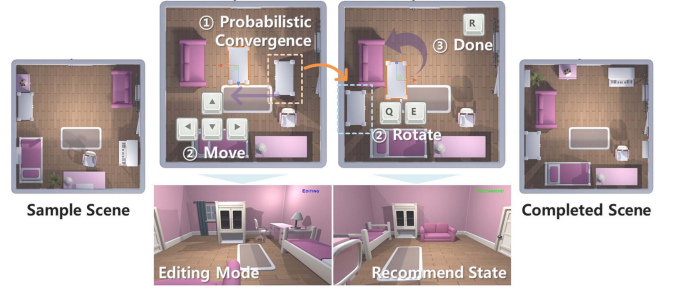


Fig. 5. An example of the editing process from probabilistic convergence.

When the object converges probabilistically, the ongoing arrangement process is broken, and the editing mode is switched. In editing mode, the user moves the object by pressing the left/right/up/down keys and rotates the object by pressing the Q/E keys. Once the edited object reaches the ideal position and orientation based on the memoized exploration array, a recommendation is provided, regardless of its alignment with the suggested area. After editing is complete, automatic arrangement mode switches to manual input (R-key), resuming the dynamic programming-based arrangement process.

V. APPLICATION

The proposed dynamic-programming-based object arrangement method relies on sample scenes, similar to existing optimal solution techniques. Machine learning application methods utilizing generative models, such as variational autoencoder, generative adversarial network, or reinforcement learning, also face limitations due to their heavy reliance on learning data. This study aims to propose a new approach to mitigate the reliance on sample scenes by automatically generating samples from real-world indoor scene data. By extracting information about the arrangement of objects in real indoor scenes, we conduct applied research on scene synthesis based on digital twins.

By introducing a new paradigm for interior space design and reconstruction, the aim is to broaden the applicability across various fields related to indoor space structures, including architecture, interior design, home styling, and real estate.

A. Sample Scene Based on Reality

The proposed approach for calculating sample scenes is based on data classified and inferred from reality. This approach is intended to be used not only in virtual spaces but also in real world scenarios. Through this process, scenes and situations that occur in reality can be edited and simulated on a computer using a digital twin. Additionally, the development of a collaborative XR environment, where MR and VR users participate together, should be considered. Specifically, this study aims to establish a framework for constructing virtual scenes from reality, building upon the research utilized [36], [41] (Fig. 6).

The scene synthesis framework proposed by Na and Kim [41] consists of a geometric analysis of real space and a semantic interpretation of real objects. First, a geometric analysis of the real-world scene was performed using the Microsoft

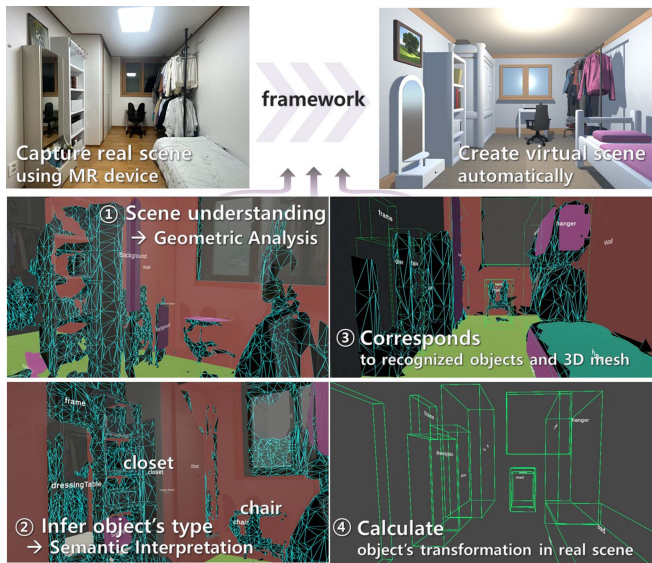


Fig. 6. A framework for calculating sample scenes based on reality.

HoloLens 2 MR device and the scene-understanding SDK. A scene representing the composition of individual objects (e.g., walls/ceilings/floors), referred to as SceneObject instances, was generated. Using the scene understanding SDK and Mixed Reality Toolkit, the 3D mesh information constituting the scene via Microsoft HoloLens 2 was classified according to predefined instances (SceneObjectKind). The following is a semantic interpretation of the objects that constitute real space. The objects constituting the scene were recognized and classified using Azure Custom Vision. The image dataset used for training the classification model was defined by selecting images with visual variety (camera angle, lighting, etc.), tagged with the objects to be classified. Using the training data, images captured through the Microsoft HoloLens 2 device were linked, allowing for the precise inference of objects in the current scene along with their associated tags. Through these processes, a labeling task was conducted to correlate the 3D object information (sceneObject) and the recognized and classified object information (tag) with the structure of the layout, extracting the types of objects and their 3D transformations (position, orientation, and size). Finally, based on the extracted object information, predefined virtual object models were automatically arranged to generate a sample scene.

Once the layout that composes the scene is provided and the bounding surfaces for the basic properties of all objects are defined, the prior of Top-Down cost and the pairwise relationship, along with the accessibility of Bottom-Up cost, are automatically calculated within the dynamic programming process. Therefore, the memoization and tabulation calculations from the generated sample scene are the same as those of existing methods. However, since the pairwise relationship can reflect the user's intent, it has been made possible for direct input. In this study, this was specified manually because the real scene was intended to have a pairwise relationship between the desk and the chair.

B. Rearrangement

Once a sample scene was created based on reality, the object arrangement process applied the proposed method based on dynamic programming. The scene synthesis result is created using the calculated virtual scene and the objects arranged based on it. This was used in a survey experiment for experts (Fig. 9).

VI. EXPERIMENTAL RESULTS AND ANALYSIS

The indoor scene synthesis experiment, results, and development environment construction using the proposed dynamic programming were implemented using C# script language based on Unity 2022.3.6f1 (64 b) [42]. The PC used for the development environment and experiment was equipped with an AMD Ryzen9 (Zen4) 7945HX (2.5 Hz), 32 GB of Ram, and a Geforce GTX 4070 Laptop GPU.

A. Overall Analysis of the Proposed Method

The first experiment is a comparison of qualitative results, key performance, and time efficiency for scene synthesis rearranged using the proposed method. Fig. 7 shows the indoor scene synthesis results generated using the proposed method. It illustrates the calculation of the natural arrangement results across a total of eight different spaces as backgrounds. The graphic resources [43], [44], [45], [46] used in the experiment were indoor scenes purchased from asset stores. Additionally, the default demo scene provided by each asset was used as a sample scene to minimize the time required to produce the sample scene.

This study employed dynamic programming as a new method for fast and accurate object arrangement. Table I summarizes the quantitative experimental results obtained using the proposed method. The experiment recorded the accuracy and time according to the threshold of the convergence condition in the process of generating scene synthesis results for a total of eight different configurations. The experiment was performed through repeated synthesis of approximately 45–55 batches per scene. Currently, the criteria for judging the accuracy (or inaccuracy) of the results are somewhat ambiguous. Among existing studies, the arrangement distribution was compared with the original, but because this study attempts to include various (unexpected) results, it is difficult to determine the accuracy only by comparison with the original. Additionally, naturalness (or aesthetic) also reflects a subjective experience, making it difficult to set a single absolute criterion.

In this study, inaccuracy was defined as objects being arbitrarily arranged in locations based on probability rather than following defined samples and relationships. Accuracy was measured by the number of objects for which the arrangement converged, excluding fixed objects, as per the third condition. Results were considered accurate if 10% or less objects constituting the scene converged according to the third condition. However, it is worth noting that even if an arrangement converged based on probability, it might still yield unnatural outcomes, such as objects being arranged in the center of the scene or losing pairwise relationships. Nonetheless, not all results necessarily led to unnatural



Fig. 7. Examples of indoor scene synthesis generated through the proposed dynamic programming-based object arrangement (left to right: bathroom, bedroom, kitchen, living room, lobby, kindergarten, coffee shop, and museum); (a) sample scene, (b) results of arrangement, (c) generated scene.

or awkward scenes. Given the challenge of defining absolute criteria for accuracy, this study evaluated accuracy based on the following criteria (probability conditions).

Through repeated experiments, we inferred a threshold value with an ideal relationship between time and accuracy. Based on this, the average time and accuracy were analyzed according

to random and priority-based selections in the selection step. It was confirmed that selecting small objects first in a random selection increased both the arrangement time of large objects and the frequency of probabilistic convergence increased.

Analysis of the experimental results revealed that, in general, the more objects involved, the longer it takes to complete

TABLE I
PERFORMANCE EXPERIMENT (TIME, ACCURACY) RESULTS OF THE PROPOSED DYNAMIC PROGRAMMING-BASED OBJECT ARRANGEMENT METHOD

	number of objects		pairwise relationships	object selection	average time (s)	accuracy (%)
	arrange	fixed				
bathroom	12	3	none	random priority	2.12 1.01	91.81 100.0
bedroom	11	4	student desk - chair	random priority	4.82 3.16	77.98 96.89
kitchen	15	5	kitchen table - stool (3 units)	random priority	4.91 3.17	95.79 99.48
living room	20	7	couch - TV couch - cabinet	random priority	5.21 4.15	98.79 99.18
lobby	25	11	couch - TV - table piano - piano seat platform	random	15.76	97.20
			- stool (3 units) office table - chair painting table - tripod	priority	18.44	91.84
kindergarten	26	9	kids table - chair (4 units) $\times$ 2set	random priority	10.19 9.11	97.79 97.72
coffee shop	24	15	table - chair (4 units) $\times$ 3set	random priority	31.17 29.98	88.89 89.74
museum	23	8	none	random priority	8.47 6.52	98.19 98.69

the process. However, a key finding is that the presence of a greater number of pairwise relationships significantly increases processing time. This is attributed to the complex and numerous conditions required for the convergence of child objects in a pairwise relationship, compared to other objects. Specifically, while rotations of other objects are fixed in four directions, child objects undergo calculations in small units, leading to a longer convergence time. Therefore, the presence of pairwise objects, or a high number of pairwise object combinations, increases the time required to complete the task. This is partly due to the need for increased accuracy. However, the size of the space (or layout) has a greater impact on accuracy. Even if the scenes have a similar number of objects, differences may arise between scenes with many large objects in a small space and those with many large objects in a wide area. For example, when comparing a bedroom and kitchen, despite having the same number of pairwise relationships, the kitchen, with more objects, still had a similar processing time. The main reason for this was that the kitchen spaces were relatively large.

As confirmed through the experiments, the resulting arrangement time depends on the pairwise relationship (particularly the number of child objects), and the accuracy depends on the size of the space relative to the number of objects (including object sizes). One thing to note is the order of the arrangement. Generally, random selection results in lower accuracy and efficiency, particularly when objects are selected in order from small to large. However, exceptional results were obtained for lobbying. This applies to cases where objects are randomly selected from a relatively large space, although there are many pairwise relationships. When the exploration range of an object considers the entire space, some objects occupy the area in advance when a random selection is performed. In this case, the

TABLE II
PERFORMANCE COMPARATIVE EXPERIMENT RESULTS WITH EXISTING RESEARCH [9] BASED ON OPTIMAL SOLUTION TECHNIQUES

	comparison	time (s)
bedroom	simulated annealing	89.81
	Our	3.16
living room	simulated annealing	152.64
	Our	4.15
museum	simulated annealing	176.72
	Our	6.52

exploration area of the later selected objects (particularly child objects of pairwise relationships) was reduced, which shortened the arrangement time and improved accuracy. This study derived an ideal idea through experiments, in which random selection can be appropriately applied to scene synthesis with complex relationships in a wide space to increase efficiency.

The following presents a comparative experiment with the object arrangement method based on the existing optimal solution technique. A comparative experiment was conducted with the optimized object arrangement through simulated annealing proposed by Yu et al. [9]. The scenes used for the comparative experiment are the bedroom, living room, and museum in Fig. 7. Table II shows the comparative experimental results. This was confirmed that there was a large difference in the time required to arrange objects. Compared to inferring all objects constituting a scene using a single cost function, it was proven that the process of solving the arrangement problem sequentially on an object-by-object unit is much more efficient. Accuracy could not be measured because there was no standard for direct comparison. However, when recording the number of overlaps (accessibility) between objects or unintended arrangement (closet arranged in



Fig. 8. Experimental results to verify the diversity of the proposed method: (a) bathroom, (b) bedroom, (c) kitchen, (d) living room.

the center of the scene), it was confirmed that the existing method showed an accuracy of about 50 to 70%.

Finally, this is an experiment on the diversity of the scene rearrangement results of the proposed method. This study basically rearranges scenes with fixed objects by category. For this reason, the rearrangement results maintain diversity, but the diversity in the scene is inevitably limited. Therefore, we conduct an experiment to verify the possibility of the diversity of the proposed dynamic programming-based method. Fig. 8 shows the results of the diversity experiment. Multiple objects are registered by category, and after the rearrangement is completed, objects are randomly selected from the category to complete the scene. As a result, we can confirm the results of various directing even with the same rearrangement results. Through this, we verified the possibility of synthesizing scenes with various directing and concepts in the same scene by adding a process of defining multiple objects by category in advance.

B. Evaluation of Editing Method

The next experiment analyzed the proposed editing function. When switching to the editing mode through probabilistic convergence, the user evaluates the scene based on the objects arranged thus far and proceeds with editing. Users can obtain recommendations from the computed cost function using the prior relationship during the editing process, and arrange them in the desired location (considering free space, furniture manipulation, and aesthetic factors). However, as there is no limit to the editing time that reflects the user's intention, this study measured the editing time up to the recommended location. The object arrangement was based on priority-based selection, and time was measured in the bedroom, living room, and lobby, where the probabilistic convergence frequency was relatively high. Table III shows this. The time required to edit an object is approximately 2~3s, which is proportional to the size of the space. This is similar to the average time required to synthesize a scene, but it can be considered efficient compared to the manual operation of the entire scene. In addition, when scene

TABLE III
ANALYSIS OF THE TIME REQUIRED FOR EDITING

	the number of editing objects	average time (s)	intervention rate
bedroom	1~2	4.013	2.7%
living room	2~3	5.171	3.2%
lobby	3~4	9.712	8.1%

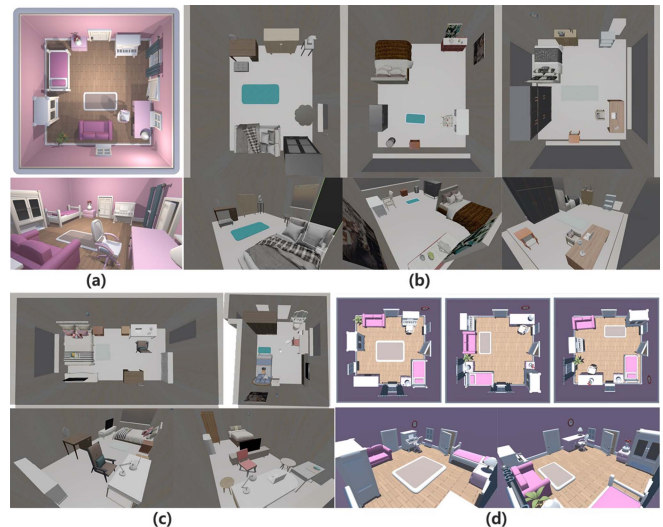


Fig. 9. Results of comparative experiments with existing interactive editing techniques and tools: (a) reference scene, (b) result produced with the same layout as the reference scene, (c) result of freely producing the scene using Zhang et al. [14], [15]'s interactive tool, (d) result of freely producing the scene using Unity 3D [42].

rearrangement was performed 100 times each for a total of 3 times, the recorded intervention rate was found to be less than 10%. This indicates that accurate scenes can be induced primarily through dynamic programming alone. Furthermore, users can re-edit scenes to reflect their intentions through direct intervention.

The following is a comparative experiment with Zhang et al. [14], [15]'s interactive editing technique, MageAdd, and SceneDirector. It is an interactive tool that intelligently infers objects in a scene by incorporating and simulating user preferences with minimal input. It also provides users with intuitive control over how to arrange groups of objects and where to arrange them. Fig. 9 shows the results of the comparative experiment. First, we measured the time required to create the same scene (Fig. 9(b)) using one of the bedroom scenes produced based on the proposed dynamic programming method as a reference (Fig. 9(a)). Next, we freely created natural scenes in the desired style using the interactive tool of Zhang et al. [14], [15] and Unity 3D, a 3D interactive engine (Fig. 9(c) and (d)). The experiment was conducted by three researchers who conducted this study. This was to compare and analyze the characteristics of existing interactive tools based on a high level of understanding of this study. Table IV shows the time taken during the production process. First, since there was no understanding of existing interactive

TABLE IV
COMPARATIVE ANALYSIS OF PERFORMANCE WITH EXISTING INTERACTIVE
EDITING TECHNOLOGIES AND TOOLS

comparison		average time (s)
reference scene	tool learning	86.67
	creation	295.67
natural scene	Zhang et al. [14], [15]	246.33
	Unity 3D [42]	276.00

tools, we measured the time taken to learn the tool. And the time taken to create the reference scene was measured to be 295.67 s on average. Considering the editing time, it took a lot of time compared to this study, which took an average of 12~15 s to create a bedroom. However, it may not be able to satisfy the advantage of existing studies in that they can freely reflect users' preferences, thoughts, and aesthetic factors. In addition, based on the tendencies of experimenters who have high utilization of the Unity 3D engine, we conducted a comparative experiment with the interactive tool of Zhang et al. [14], [15]. As a result, we were able to confirm that the interactive tool of Zhang et al. [14], [15] showed slightly better performance. Therefore, if the proposed method appropriately reflects the advantages of the interactive structure of existing studies, it is judged that efficient scene rearrangement will be possible while applying the aesthetic factors, interaction, and user preferences of the generated scene.

C. Comparison With State-of-The-Art Methods

In an additional quantitative experiment, we measured the plausibility of the synthesized scenes using scene classification accuracy (SCA). A score close to 50% indicates that the generated scene is indistinguishable from the real scene. SCA is calculated by classifying the rendered images of the synthesized scene and the rendered images of the real scene. In contrast to existing deep learning-based studies [11], [28], [34], this research has a structure that calculates sequentially from a single sample without synthesizing scene using the 3D-FRONT dataset [47]. Therefore, the real scene image is composed by rendering a single sample scene from multiple angles and mixing real images by category. The rendered image is also composed by mixing the image generated using the proposed method with an illustration image or a scene processed using Blender. In other words, the classification was performed based on an extended concept, not just SCA limited to the 3D-FRONT dataset. Since the proposed method and existing studies have different purposes and conditions, the comparison was performed by referring to the performance results of the existing study [28] for objective comparison. Table V presents a quantitative comparison based on SCA evaluation metrics. The proposed method demonstrates comparable performance to previous studies. In addition, we conducted experiments across a broader range of scene categories than those in previous studies and observed similar results. Therefore, the proposed method demonstrates its ability to generate plausible scenes while considering a variety of scenes.

Next, a difference comparison experiment was conducted with deep learning research. Recently, research applying deep learning for indoor scene synthesis has been actively explored. Various studies aim to achieve accurate and efficient scene synthesis by considering the interactions, movement lines, and motion in the virtual environment [13], [31]. However, due to the nature of deep learning-based approaches, direct comparison and analysis with this study, based on optimal solutions, are challenging. Therefore, the differences from previous studies were compared by dividing them into models, datasets, objectives, and time.

Table VI shows this. Since the SUNCG dataset [48] is currently unavailable, we compared the differences with representative and recent studies based on the 3D-FRONT dataset [47]. Existing deep learning studies are characterized by their ability to naturally generate diverse scenes from a wide range of categories and object sets. However, these studies have several limitations. For instance, while HAISOR [13] incorporates human awareness, it demands extensive preliminary tasks compared to predecessors like ATISS [11]. Similarly, the static generation process of Transformer [49] lacks any mechanism for user interaction or editing. Recent approaches, such as InstructScene [28], are limited to generating small-scale scenes only. In addition to the limitations of individual models, there is a fundamental issue with the widely used 3D-FRONT dataset. Even after filtering, it is known to contain problematic object arrangements and misclassifications. Furthermore, existing studies have a high dependency on the 3D-FRONT dataset, to ensure there are clear limitations in considering applications or expanding to scenes with various concepts. In addition, despite the advantages of learning, the local minima problem still exists. In contrast, the proposed method converts the local minima problem into probabilistic convergence and reflects the user's intention through an editable method to consider cognitive or aesthetic factors. This process may increase generation time, but it distinguishes itself from existing studies by allowing interaction and editing during the arrangement process. In addition, low dataset dependency can also be advantageous from an application perspective. Nevertheless, the diversity of scene synthesis results is an inevitable weakness because scenes are generated only with specified objects. However, Song et al. [49] perform learning with specified objects through a fine-grained optimization process to address local minima issues, such as object collisions. While generating scenes with only specified objects is a significant issue that requires improvement, it is not a limitation unique to the proposed method.

Furthermore, Yang et al. [50] conducted scene synthesis studies. The proposed method combines the advantages of each methodology by considering optimization methods in a deep learning-based learning model. It learned latent code using 3D scenes as input, and confirmed that various scene synthesis is possible with the learned vectors. Therefore, it was confirmed that complete 3D scene synthesis is possible using incomplete 3D scenes as input. In addition, it has the characteristic of being able to output natural and diverse arrangement results by reflecting some randomness in the learning model by creating a Bayesian optimization-based relationship. However, it also has

TABLE V
QUANTITATIVE EVALUATION OF SCENE SYNTHESIS RESULTS BY THE PROPOSED METHOD FOR ROOM TYPES

SCA%	bedroom				living room			
	Ours	ATISS [11]	DiffuScene [34]	InstructScene [28]	Ours	ATISS [11]	DiffuScene [34]	InstructScene [28]
	60.41 $\pm$ 4.13	59.17 $\pm$ 1.39	60.49 $\pm$ 2.96	56.02 $\pm$ 1.43	65.41 $\pm$ 3.57	69.38 $\pm$ 3.38	72.92 $\pm$ 1.29	65.42 $\pm$ 2.52
	bathroom		kitchen		kindergarten		coffee shop	
	59.13 $\pm$ 3.84		60.03 $\pm$ 3.90		61.98 $\pm$ 4.39		62.51 $\pm$ 3.12	

TABLE VI
COMPARATIVE ANALYSIS OF THE DIFFERENCES BETWEEN THE PROPOSED METHOD AND EXISTING DEEP-LEARNING BASED STUDIES

Method	ATISS [11]	HAISOR [13]	Non-autoregressive Transformer [49]	InstructScene [28]	Ours
Model	Autoregressive Transformer	GNN + RL (MCTS)	Non-autoregressive Transformer	Semantic Graph Prior + Layout Decoder	Dynamic Programming
Dataset	3D-FRONT (~12K scenes)	3D-FRONT (2 scene types)	3D-FRONT (~5K scenes, augmented)	3D-FRONT (~6K scenes)	N/A (no learning required, only 1 sample)
Categories	bedroom, living room, dining room, library	bedroom, living room	bedroom, living room	bedroom, living room, dining room	bedroom, bathroom, kitchen, living room, lobby, kindergarten, coffee shop, museum
Main Objective	unordered, automatic, and object suggestion	human awareness	fine-grained optimization of collisions	instruction-driven synthesis	editable & applicable rearrangement
Collision	partial	improved	improved (complex collision problems exist)	controllable	complete (prevented)
Local Minima	persists	persists	not addressed	controllable (application of LLM)	solved
Generation Time	88.24 ~ 102.38ms	not specified	approximately 8h (training), 18.91s (inference)	about 12s (living room)	1.01 ~ 31.17s (depending on the number of objects)

limitations in that pre-learning is required, user intervention is impossible, and consistency is low. Moreover, since the relationships between objects are not explicitly enforced, the results for pairwise relationships between objects may not be accurate or consistent.

This study extracts the information required for scene arrangement from a minimum number of sample scenes through an optimization method based on dynamic programming and calculates the optimal result through repeated operations at the object level. We are also considering a function that can reflect the user's intention without abandoning the incomplete scene synthesis results by introducing probabilistic convergence and editing processes during the arrangement process. Therefore, if a user needs a scene that matches the desired situation and production, it has the advantage of being able to quickly derive the desired result by defining only a sample scene without building a dataset and performing learning.

In addition, we qualitatively compared and analyzed the purpose, method, and features of the recently actively conducted LLM-based scene synthesis researches. Table VII summarizes the comparison results. These studies have improved the user's intention, sophisticated spatial structure analysis, and consistency through prompts. Furthermore, they have overcome some of the limitations of deep learning-based studies in scene synthesis based on datasets, and have suggested various scene creation. However, a certain level of understanding of the prompts is

required, and the quality and completeness of the scene synthesis results tend to depend on the prompt contents. Additionally, it must take into account the limitations of costs incurred in the scene creation process. Ultimately, a method is needed that can effectively generate scenes of various types and compositions while minimizing incomplete scene generation along with the process of reflecting the user's intention. The proposed dynamic programming-based method aims to effectively generate various scenes and compositions with minimal sample information and interactive structures. In addition, it was confirmed through several experiments that various types of scene synthesis are possible with a consistent structure.

D. User Study for Applicability Validation

This study aims to present not only fast and intuitive object arrangements but also usability and direction from an application perspective. For this purpose, a corresponding virtual scene was created based on a reality-based scene. In addition, new object arrangement results were generated using the proposed method. Fig. 10 shows the virtual scene created from the reality-based scene and the results of the newly arranged scene synthesis based on it. Based on this, this study formed an expert group [51], [52] consisting of interior design departments, graphic designers, design lecturers, and graphics developers in the construction and architecture fields and conducted a survey experiment. The

TABLE VII
COMPARATIVE ANALYSIS OF THE DIFFERENCES BETWEEN THE PROPOSED METHOD AND EXISTING LLM-BASED STUDIES

Comparison	
LayoutGPT [32]	Model: LLM (ChatGPT-3.5/4) + prompt Engineering.
	Main Objective: generate user-controllable 3D layouts from complex text prompts.
	Key Strengths:
	- high performance in spatial reasoning & training-free.
	Limitations & Challenges:
	- typical collision problem & post-processing required.
HOLODECK [33]	Model: LLM (ChatGPT-4) + constraint optimization.
	Main Objective: generate diverse, interactive 3D environments for AI agent training.
	Key Strengths:
	- various types of custom environments (apartments, museums, etc.) & object relationships and spatial consistency.
	Limitations & Challenges:
	- performance proportional to the complexity and constraints ($\sim 3\text{m}/\text{room}$) & direct monetary cost ($\sim \$0.2/\text{room}$).
DiffuScene [34]	Model: denoising diffusion model + text embeddings.
	Main Objective: synthesize and complete 3D scenes from text prompts, ensuring diversity.
	Key Strengths:
	- ensuring high diversity & holistic scene generation via diffusion.
	Limitations & Challenges:
	- limited to single-room generation & additional steps are required for object textures.
Ours	Model: dynamic programming.
	Main Objective: editable and applicable rearrangement with minimal user input.
	Key Strengths:
	- solves local minima via user editing and intervention.
	- facilitating user editing and intervention & consider the applicability factor.
	Limitations & Challenges:
	- generation time increases with scene complexity. & aesthetic factors are user-driven, not automated



Fig. 10. Examples of reality-based virtual scenes and rearrangement results created for the survey experiment of the user study: (a) real scenes, (b) virtual scenes created in correspondence with reality, and (c) rearrangement results generated through the proposed method.

experiment consisted of a total of nine participants (six males and three females) between the ages of 27 and 40. Participants were provided with gift cards worth approximately \$5 each. The purpose of the survey was to evaluate the efficiency, applicability, and other aspects of the proposed method based on the rearranged synthesis results. It aimed to assess whether automatically generating results similar to those in Fig. 10(c) based on one's own room (or a customer's room) from the perspectives of interior design, home styling, etc., would be effective. Therefore, user studies consist of multiple-choice surveys for statistical analysis of expert opinions based on consistent standards and subjective surveys to understand the detailed opinions of individual experts. First, for the multiple-choice survey, this

study utilized the UEQ (User Experience Questionnaire) [53], a reliable questionnaire to measure the user experience. Among all survey items, questions on attractiveness and novelty were included to fit the purpose of this study.

Fig. 11 summarizes the survey results. The obtained results for the six questions of attractiveness and four questions on novelty are shown in Fig. 11. Overall, most experts found the various synthetic results rearranged from virtual scenes created based on reality using the proposed method to be attractive. However, in some cases, without understanding the principles and structure of the rearrangement process, some felt that judging based only on the results was difficult or unfriendly. There were several similar opinions on novelty, likely because there are not many

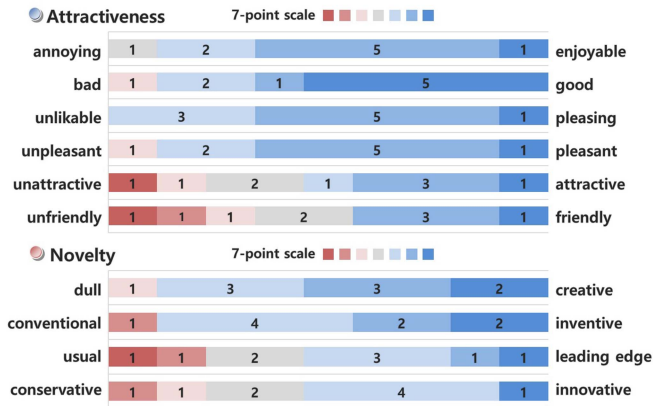


Fig. 11. Evaluation results from qualitative survey on user experience attractiveness and novelty.

TABLE VIII
SUBJECTIVE QUESTIONS TO A GROUP OF EXPERTS TO EVALUATE THE
USABILITY OF THE PROPOSED METHOD

Q1	Does it include features that will impress, excite, or motivate users?
Q2	Do the arrangement results contain practically usable results?
Q3	Among the six arrangement results, which one is the most impressive (positive)? (including reasons)
Q4	What are the most important aspects to consider when automatically arranging objects?
Q5	Would you be interested in using a system that automates object arrangement for indoor scenes such as the following?

similar existing systems. Consequently, good opinions were given in terms of novelty and originality. Nevertheless, some expressed the view that it is stereotypical and lacks innovation due to its limited arrangement and resources.

The subjective survey comprised five questions. Table VIII summarizes the questions asked in this study.

Based on the responses to the questionnaire, the contents are summarized as follows: Regarding the first question, the majority of opinions indicated that it would be a good reference for users concerned about interior design or furniture arrangements. However, it was noted that it included results that did not consider free space or movement lines for controlling objects (e.g., opening drawers). In addition, there were opinions that improvements were needed in areas where “user opinions are not reflected.” Regarding the second question, the answer was that only some of the rearranged results could be used. In particular, attention can be drawn to the opinion that “the ability to see various results quickly is considered the biggest advantage.” However, there were also many opinions regarding movement lines and questions about room size, layout, etc. Eight experts chose the third result as their answer. The reason was that “there is no blockage when opening the door and entering, and there is enough space for each furniture.” Through this, it was possible to determine that the proposed method can calculate an object arrangement that can be practically used. The next most important factor, as expected by many experts, was the space required for activities, movement lines, and aesthetic elements. In response to the last question, the eight experts were of the

opinion that it would likely be used. There were opinions such as “being able to see unexpected arrangements” and “being able to quickly check various arrangement results, making it useful as reference material.” Considering these points comprehensively, it was possible to clearly confirm positive opinions that the proposed method can serve as a good reference when applied to real-world spaces, as well as suggestions for improvement regarding user feedback and aesthetic factors. If these opinions are reflected and improved upon in the future, we believe that it will be fully applicable in industrial fields based on digital twins. Besides that, by diversifying objects by category (Fig. 8), various scene synthesis results can be utilized as reference materials for interior design and basic data for considering the interior structure of model houses in the architectural field.

VII. LIMITATION AND DISCUSSION

A. Local Minima Problem

The proposed object arrangement selects objects individually, determining their optimal positions and orientations. However, this approach can cause convergence to local minima. Prioritizing large objects in the selection criteria may lead to difficulties in placing other objects in ambiguous positions because their spaces are not considered. When these processes are sequentially accumulated, a problem arises where some objects inevitably converge into incorrect (unnatural) arrangements, such as a closet being arranged in the middle of the scene. This study proposes a new method to convert the local minima into a probabilistic convergence problem and solve it using editing functions. In addition, the possibility of reducing the probabilistic convergence through random selection was confirmed through experiments. However, the local minima (aesthetic and cognitive factors) problem remains unsolved.

B. Human Awareness

Compared to recent studies related to scene synthesis [13], which have developed methods to arrange objects by considering space for interaction and movement lines, the proposed method does not consider these factors. User evaluations targeting experts also mentioned movement lines and space for action as important factors; indicating that improvements are needed. The processing steps in the proposed Top-Down and Bottom-Up approaches are considered to have a suitable structure for addressing such issues. Specifically, by incorporating the pathway volume for movement lines and viewing frustum volume for object interactions (such as opening or closing doors) as additional Top-Down and Bottom-Up costs, respectively, as depicted in Fig. 12, these issues can be addressed effectively. Alternatively, adding user-editing functionality during the object arrangement process would sufficiently reflect aesthetic elements and user intentions. In this regard, there are plans for further enhancements and refinements through additional research in the future.

C. Verification of Various Applications

Through this application, we aimed to validate the utility of the proposed method while mitigating dependency on the

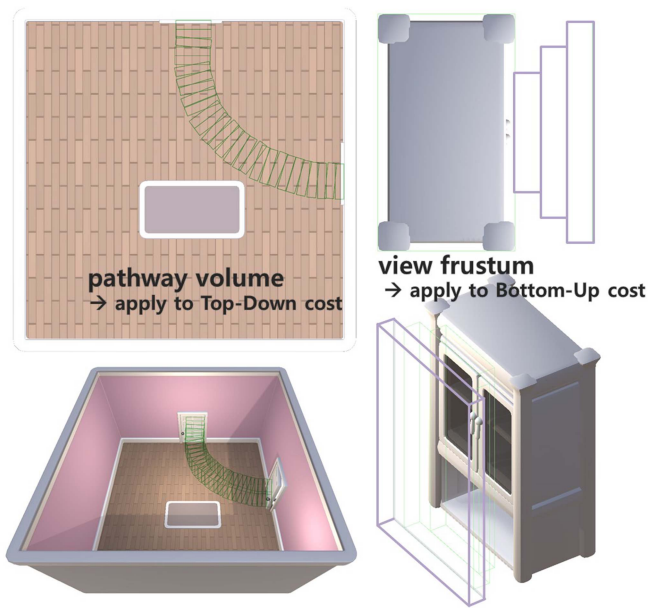


Fig. 12. Example of the Top-Down and Bottom-Up costs addition process considering movement line (pathway volume) and user action (view frustum).

samples. However, this is merely one approach, and its limitations still exist. Just as in research on indoor scene synthesis using deep learning, where dependency on training data is a limitation, overcoming the reliance on sample scenes is also a challenge. In addition, since this study aims for minimal input, only one sample scene was set. However, it is believed that if samples based on various layouts are composed in scenes of the same situation (bedroom, living room, etc.), more improved rearrangement results can be derived.

Furthermore, we conducted experiments from the perspectives of interior design and home styling, aiming to consider a broader range of applications. Additionally, by conducting comparative experiments with commercial software related to furniture arrangement [54] or by adding an interactive editing functionality to the rearranged results, we expected improved usability and enhanced utility, thus demonstrating improved results in terms of applicability.

D. Diversity of Objects

Next, unlike previous studies that randomly selected objects by category, this study performed an arrangement using only the objects specified for each scene. For this reason, there is a limitation that the diversity of scene synthesis results is low. We experimentally verified that various scene synthesis results can be derived through a structure that randomly selects multiple objects by category (Fig. 8). However, it does not take into account various layouts and irregular patterns. Therefore, more diverse scene synthesis results can be generated by developing a method that defines multiple objects within a category and selects one object from the same category by considering its size, as in an existing study using deep learning [11].

E. Depth of Quantitative Experiments

Finally, to quantitatively test the proposed dynamic programming-based scene relocation method, several quantitative experiments were conducted, including comparisons with existing studies. However, a comprehensive performance comparison with existing optimization techniques, such as greedy and genetic algorithms [21], is an issue that needs to be addressed. Also, we were unable to analyze performance metrics such as Fr chet Inception Distance (FID) and kernel inception distance, which have traditionally been used to assess the performance of deep learning-based scene synthesis. This limitation arises because, while studies based on datasets like 3D-FRONT use them as ground truth for comparative analysis, this study utilizes only a single sample scene, resulting in different conditions. As a result, making direct comparisons was challenging. Therefore, we need to develop metrics that can quantitatively analyze the performance of optimization-based scene rearrangement studies.

VIII. CONCLUSION

This study proposes an object arrangement method using dynamic programming as a new optimal solution method that is editable and applicable to intuitive structures. Generally, when arranging objects in an indoor space, each object is sequentially arranged individually from a cognitive perspective. In addition, a novel method of applying dynamic programming was designed based on the fact that the overall process of object arrangement can be analyzed by matching the overlapping subproblems and optimal substructure conditions. First, object representations are defined to interpret the attributes and relationships among objects. We then designed a Top-Down approach to interpret objects based on the overall structure of the scene (prior relationships) and predefined relationships between individual objects (pairwise relationships), as well as a Bottom-Up approach to gradually construct the overall structure of the scene through the interpretation of relationships for each object (accessibility). Accordingly, we define an optima and simple cost function that fits the designed method. Object arrangement progresses through selection, exploration, convergence, and move. The first editable factor was defined as a random and priority-based selection (with three priorities). This ensures the autonomy of object selection during the arrangement process. Next, the Top-Down and Bottom-Up costs of the current object are calculated. If the three convergence conditions defined in the process are met during movement (translation and rotation) through the cost, convergence occurs, and the arrangement is completed. A second editable factor was implemented as the user input during the arrangement process, allowing the user to intervene when probabilistic convergence led to inferred unnatural results by arranging the objects, thereby enhancing the aesthetic (or cognitive) factor. In addition, applied research was conducted to explore realistic application areas for the proposed method and confirm its feasibility. This involved generating a composite result of multiple indoor scenes rearranged using the proposed method, based on a virtual scene that corresponds elaborately to reality. We also systematically analyzed the applicability

of the generated results. Finally, to analyze the performance and usability of the new object arrangement method based on dynamic programming, scene synthesis results were produced against the backgrounds of eight diverse subject spaces. In addition, the time required and the accuracy during the object arrangement process were recorded, and the performance was analyzed. Comparative experiments or analyses were conducted with scene synthesis studies using existing optimal solution techniques [9] and applied studies using deep learning [11], [13], [49]. Various experimental results confirmed the effective performance of the object arrangement process, satisfactory scene synthesis results, and flexibility for expansion and use in various indoor scenes. Additionally, a new application direction for the proposed method was presented. Furthermore, by conducting in-depth surveys with a group of experts in the relevant field, we obtained positive responses regarding the practical usability of the proposed method. We also gathered various professional opinions regarding areas for improvement and enhancement, making the proposed method a good opportunity for exploring specific directions for future development.

In the future, based on key opinions derived from expert user studies, we plan to develop a technology that can be practically used in various industrial fields by reflecting on space, considering movement lines, actions (interaction with objects), and aesthetic elements. Additionally, by leveraging the unique aspect of object-specific arrangements in this study, we also intend to explore the development of an interactive system with user-editing capabilities. Based on this, we plan to expand the application experiments, which are currently limited to interior design and home styling, to various industrial fields, such as digital twin-based virtual manufacturing and education.

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